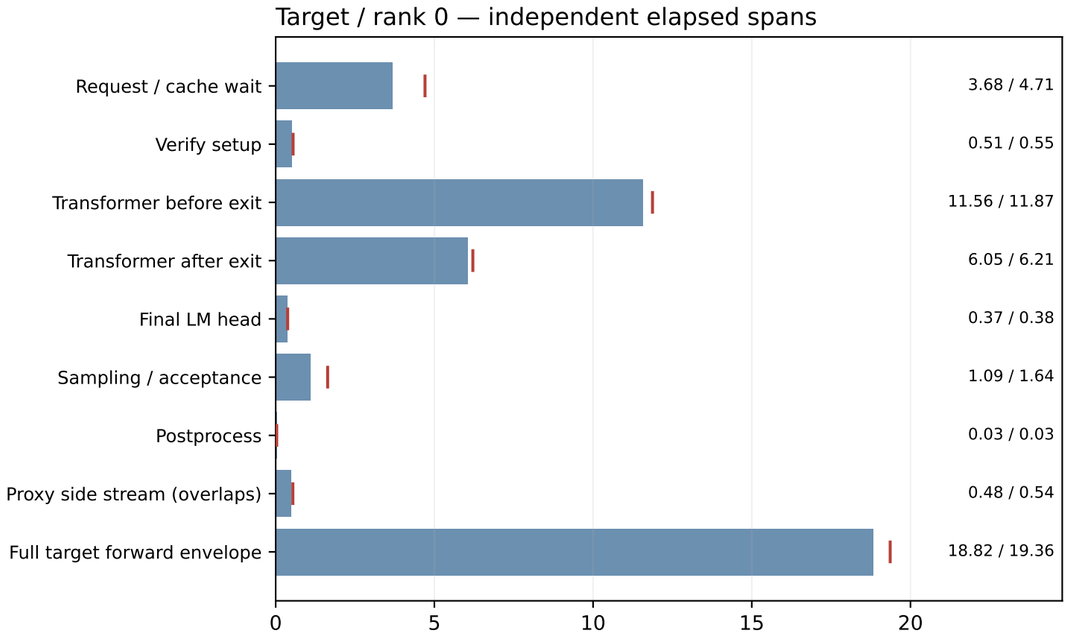
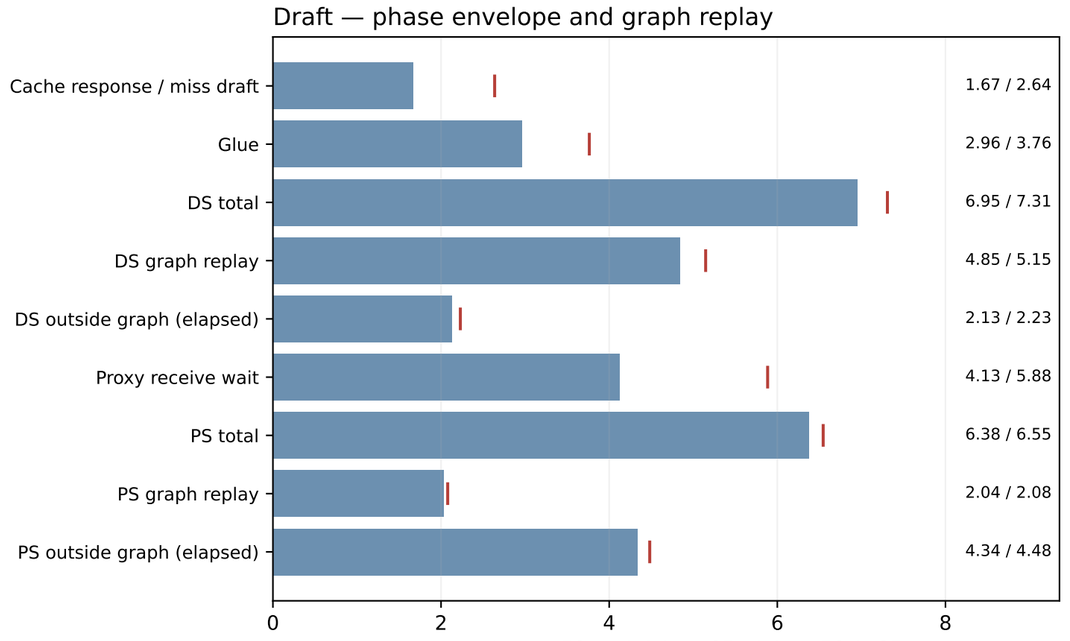
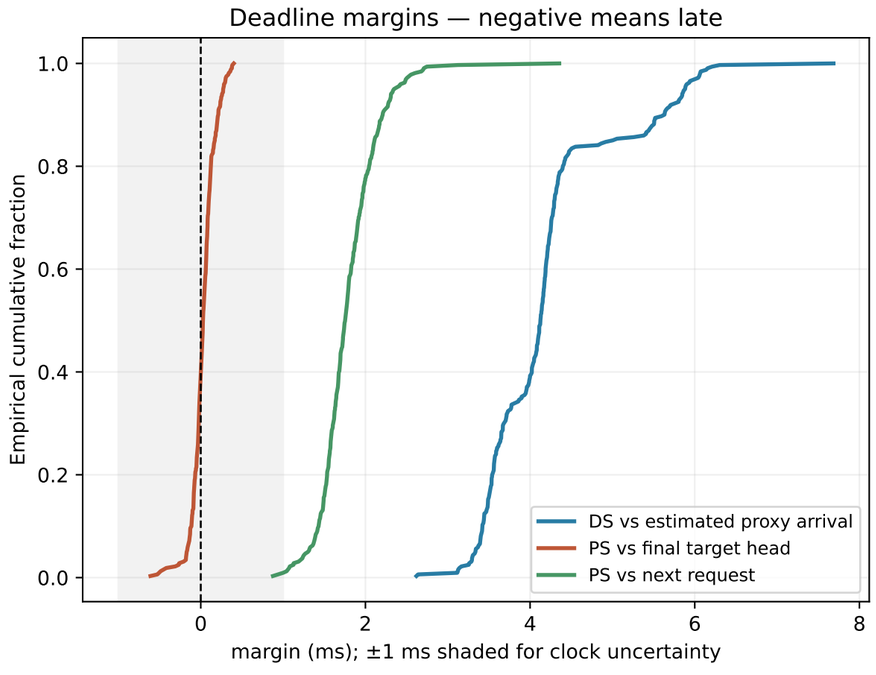
LayerSkip Llama2-7B + llama-68M
Component latency, phase deadlines, and every measured step

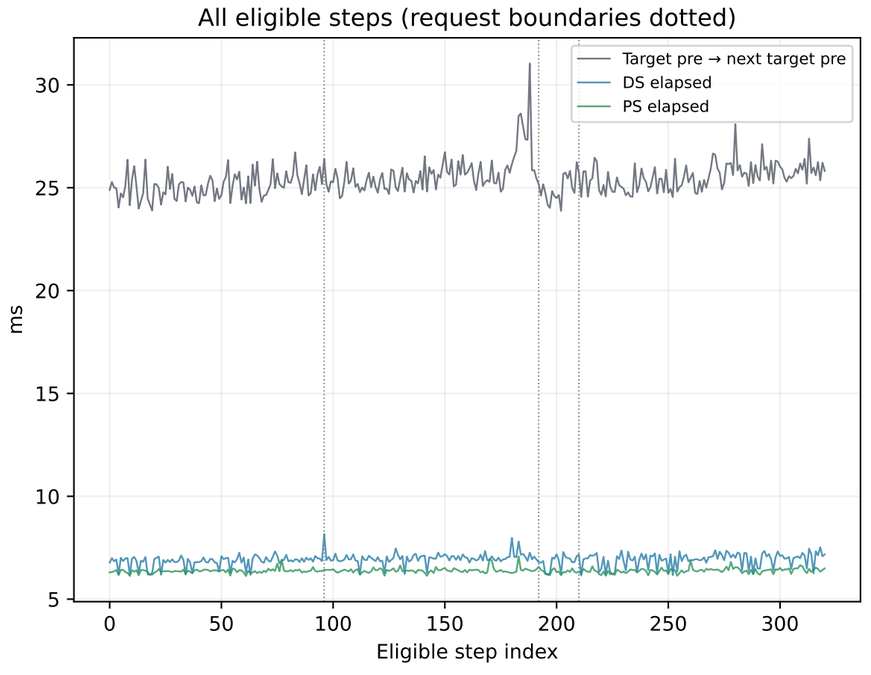
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256







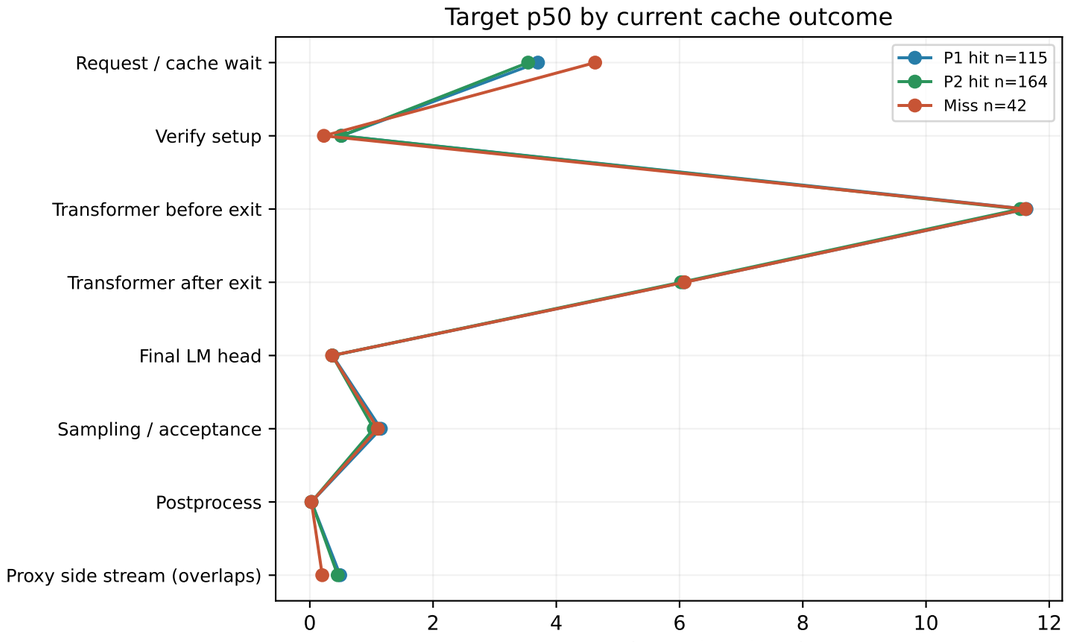


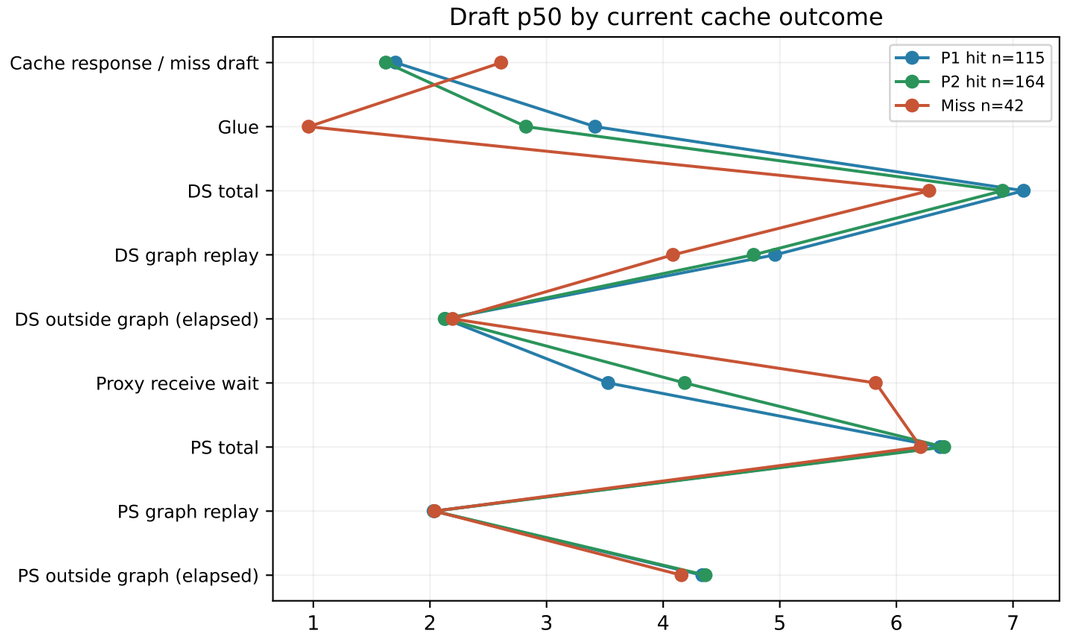
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

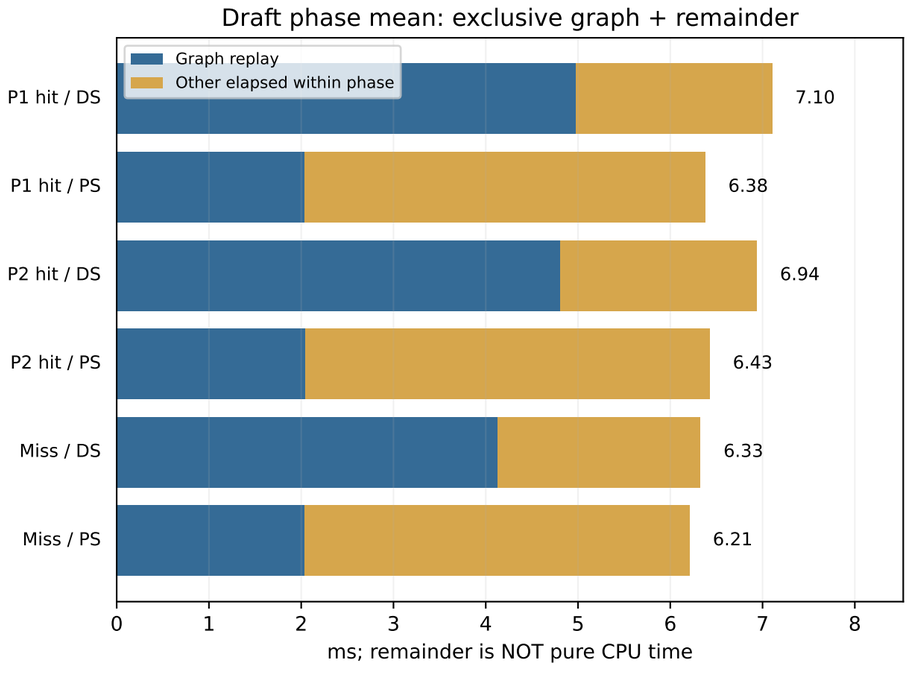
LayerSkip Llama2-7B + llama-68M
P1 hit / P2 hit / miss — component distributions and phase composition

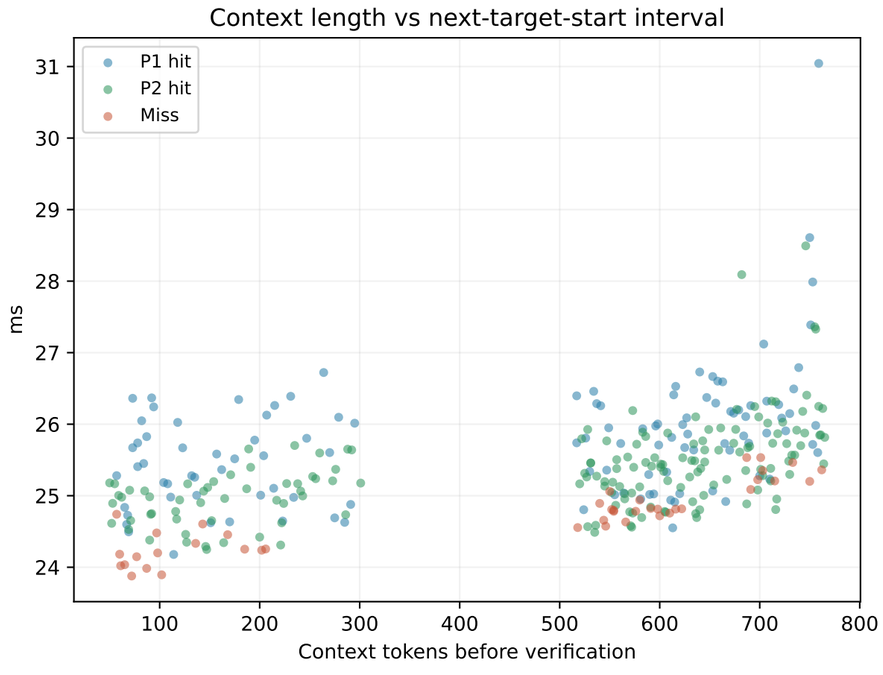
exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256









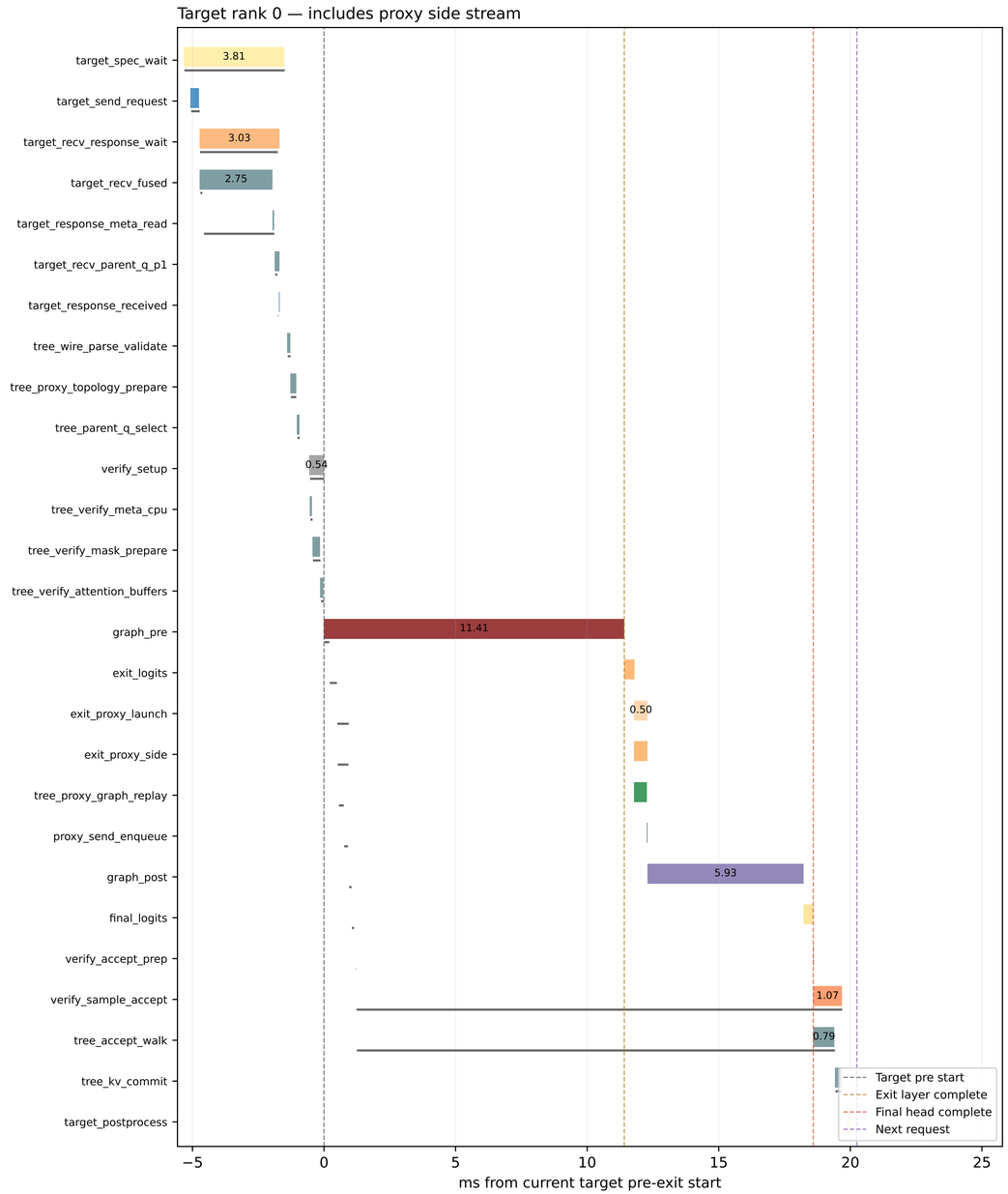
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

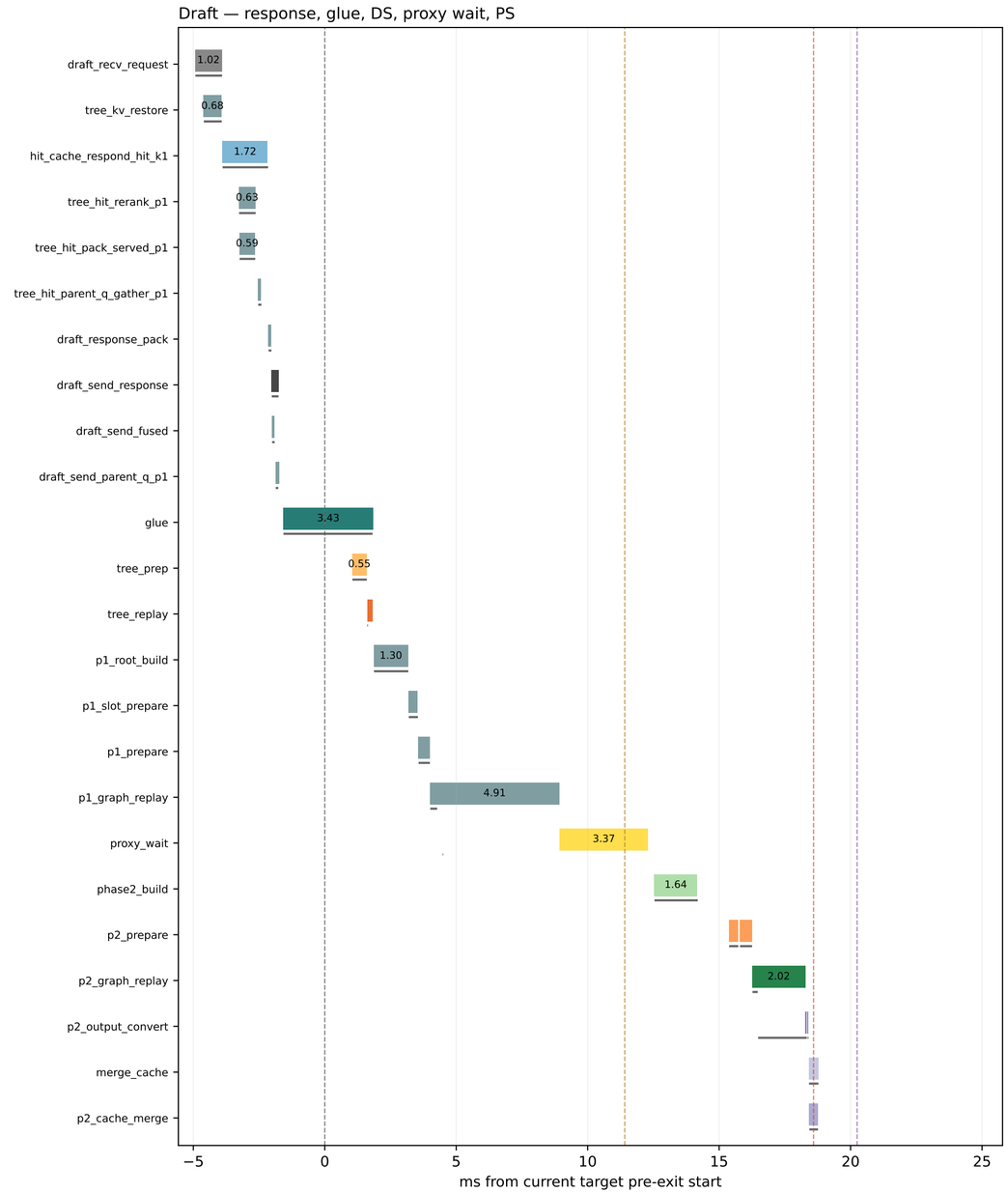
LayerSkip Llama2-7B + llama-68M
P1 hit / p25 target-step duration / request 1, UID 84_t0, step 25

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=3; DS margin +3.368 ms (estimated); PS margin -0.174 ms; next outcome P1 hit





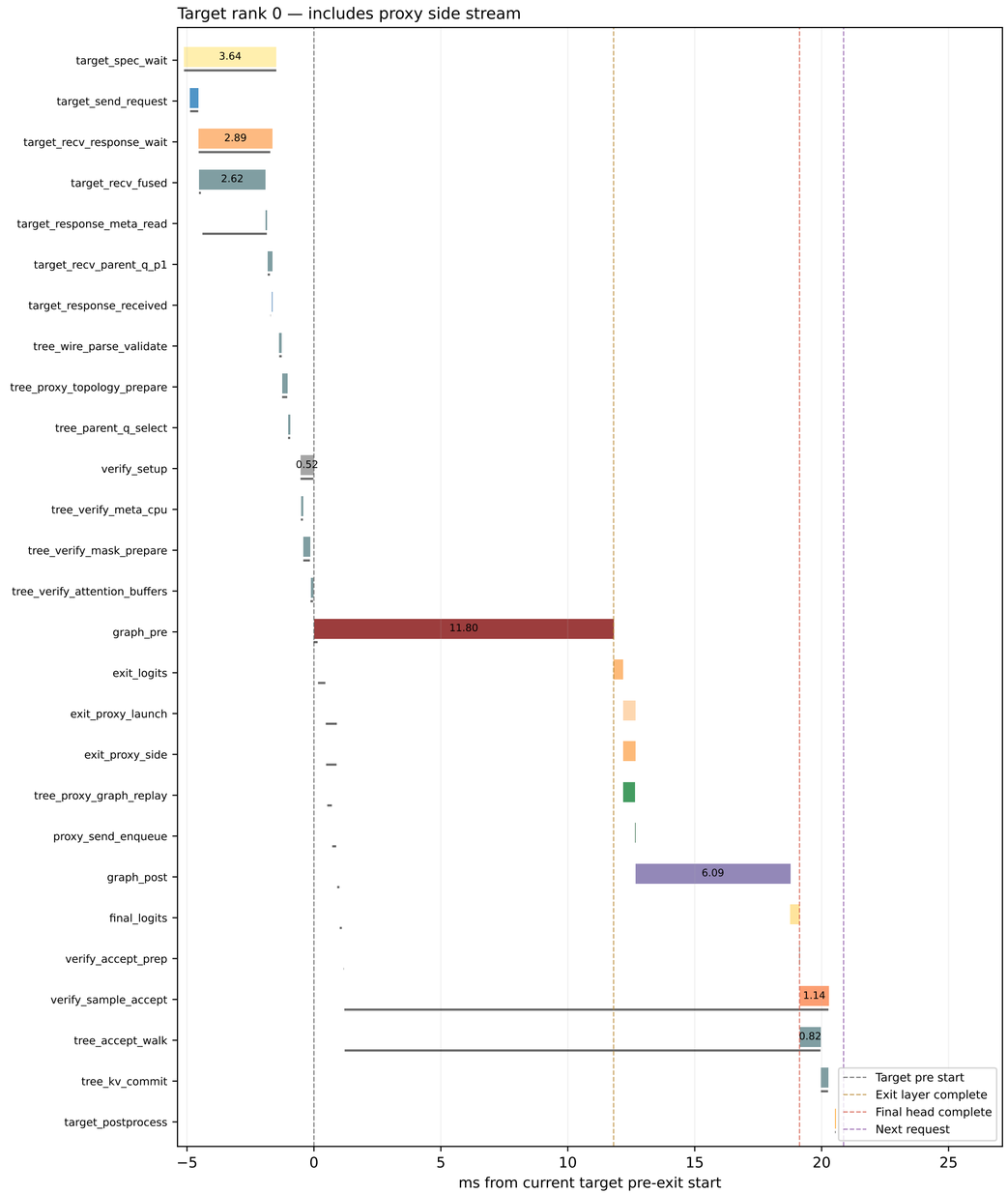
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

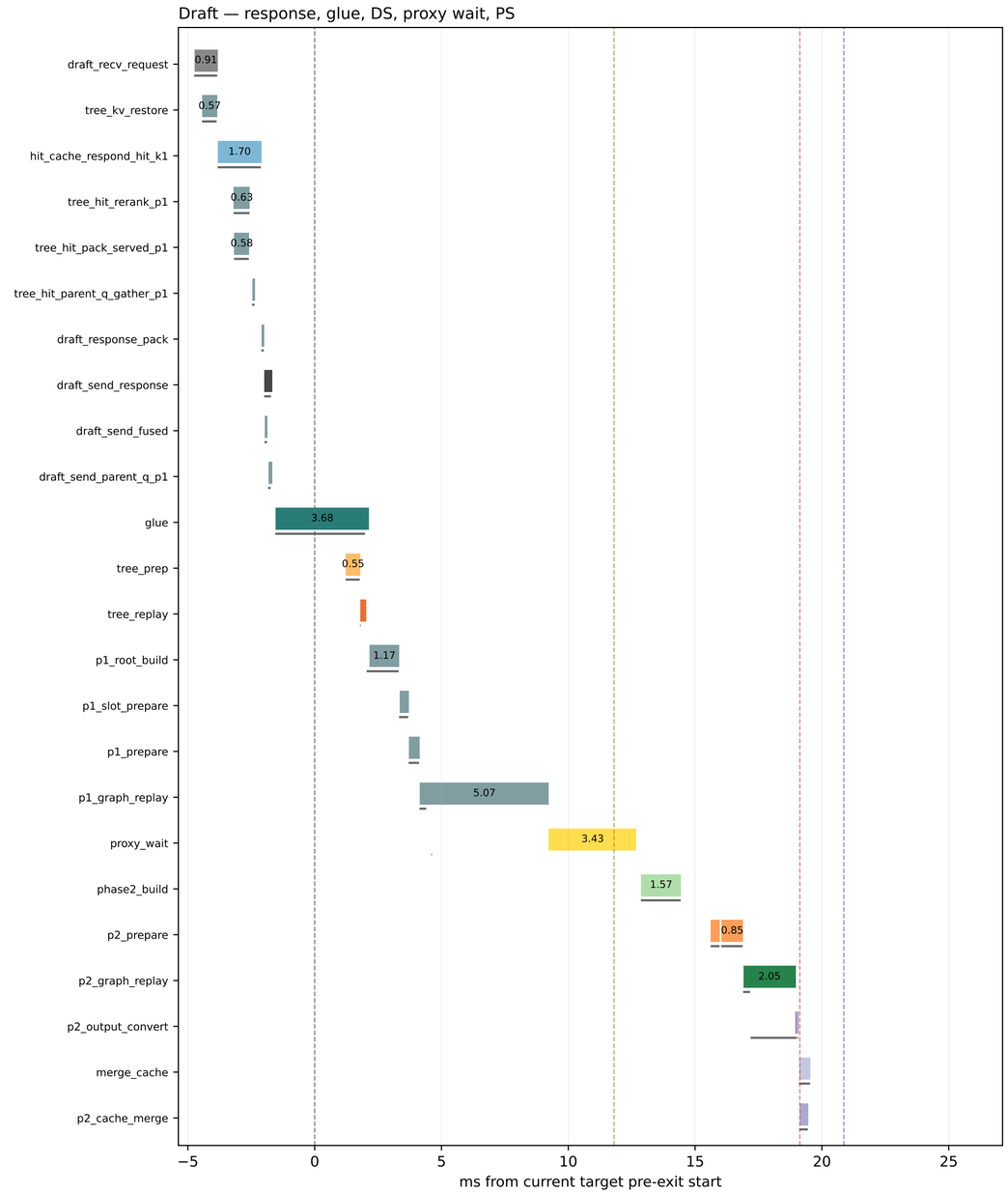
LayerSkip Llama2-7B + llama-68M
P1 hit / p50 target-step duration / request 2, UID 244_t0, step 78

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=3; DS margin +3.433 ms (estimated); PS margin -0.419 ms; next outcome P1 hit





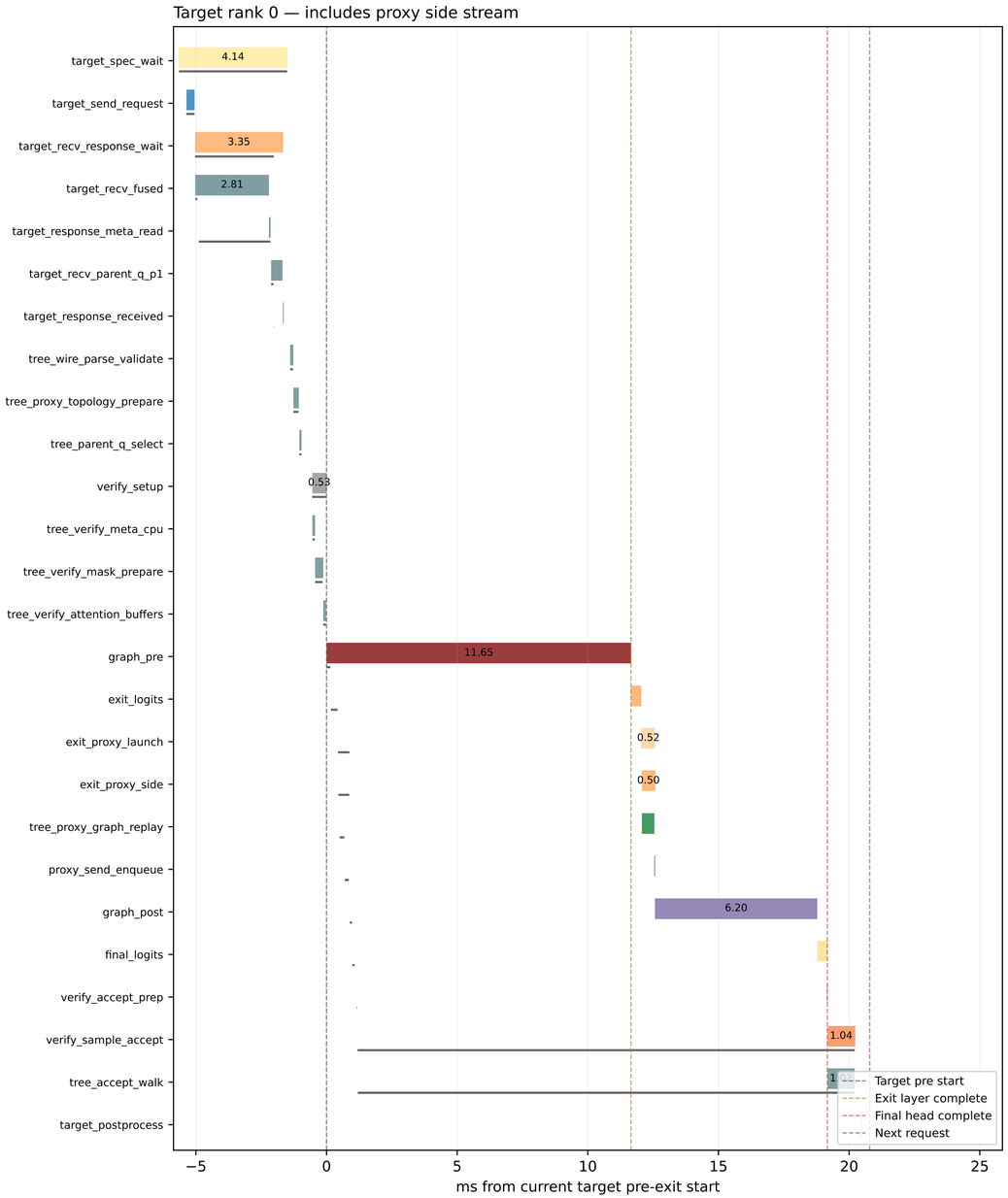
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

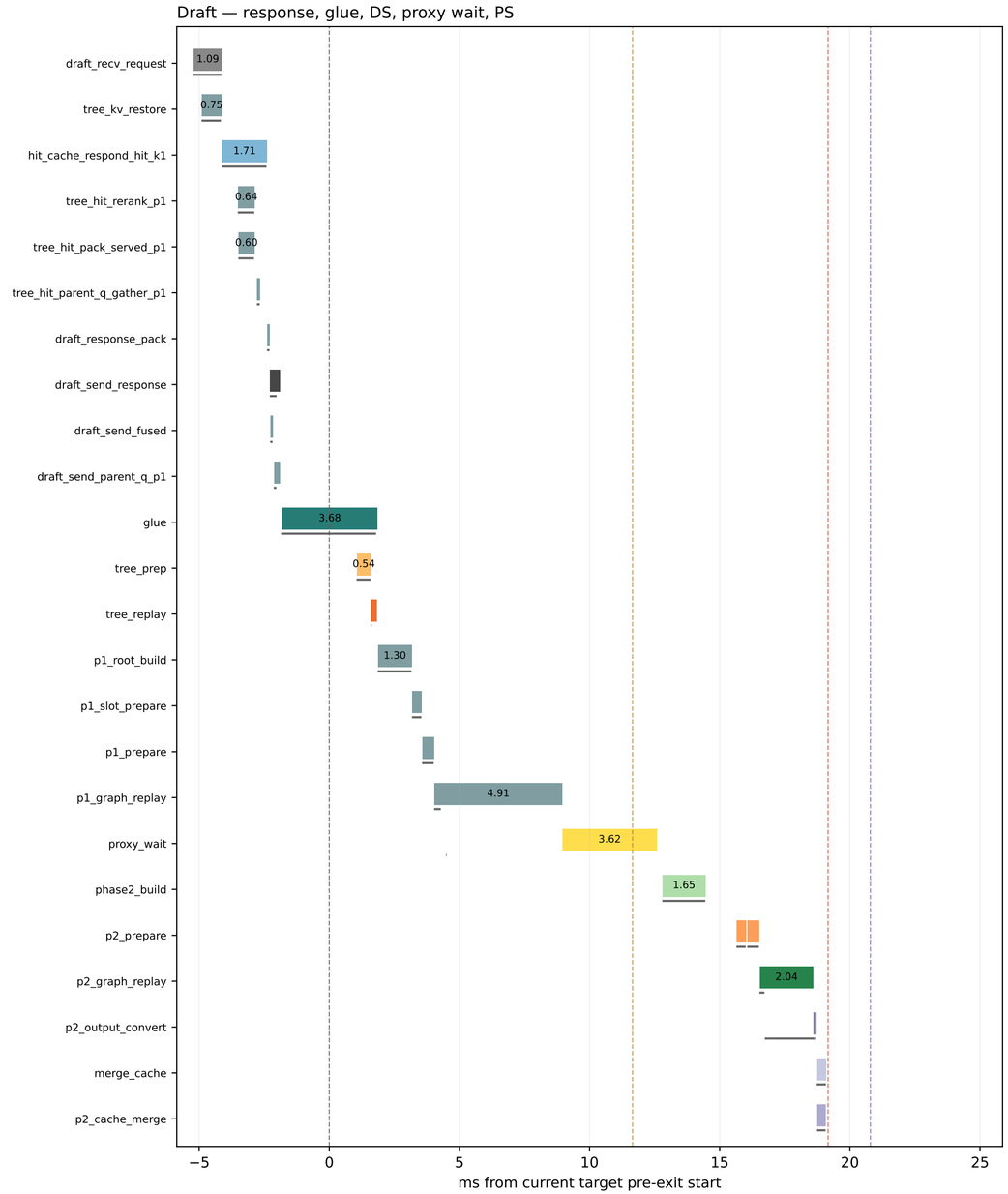
LayerSkip Llama2-7B + llama-68M
P1 hit / p75 target-step duration / request 2, UID 244_t0, step 61

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=12, AL incl. recovery=1; DS margin +3.618 ms (estimated); PS margin +0.071 ms; next outcome P2 hit





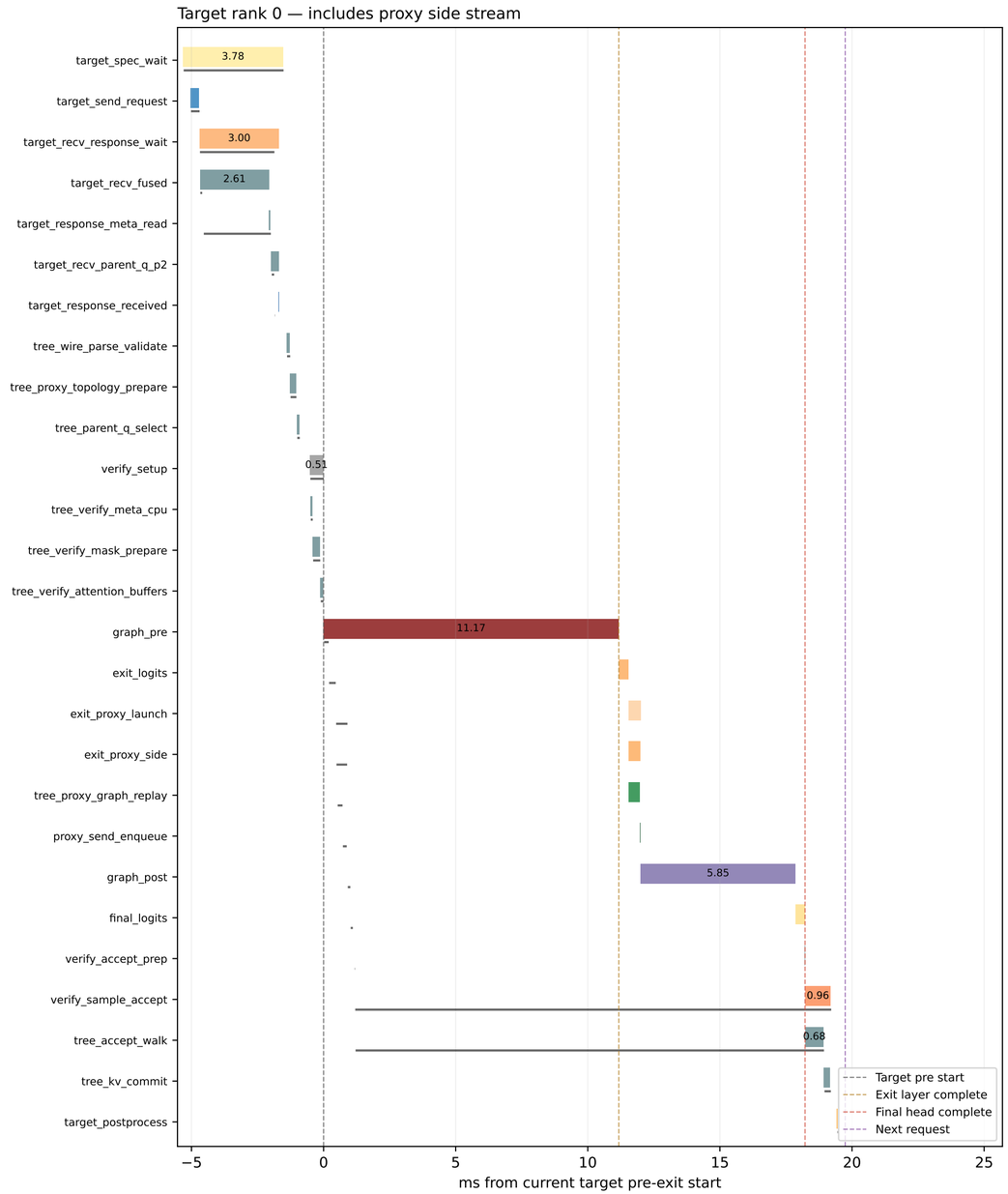
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

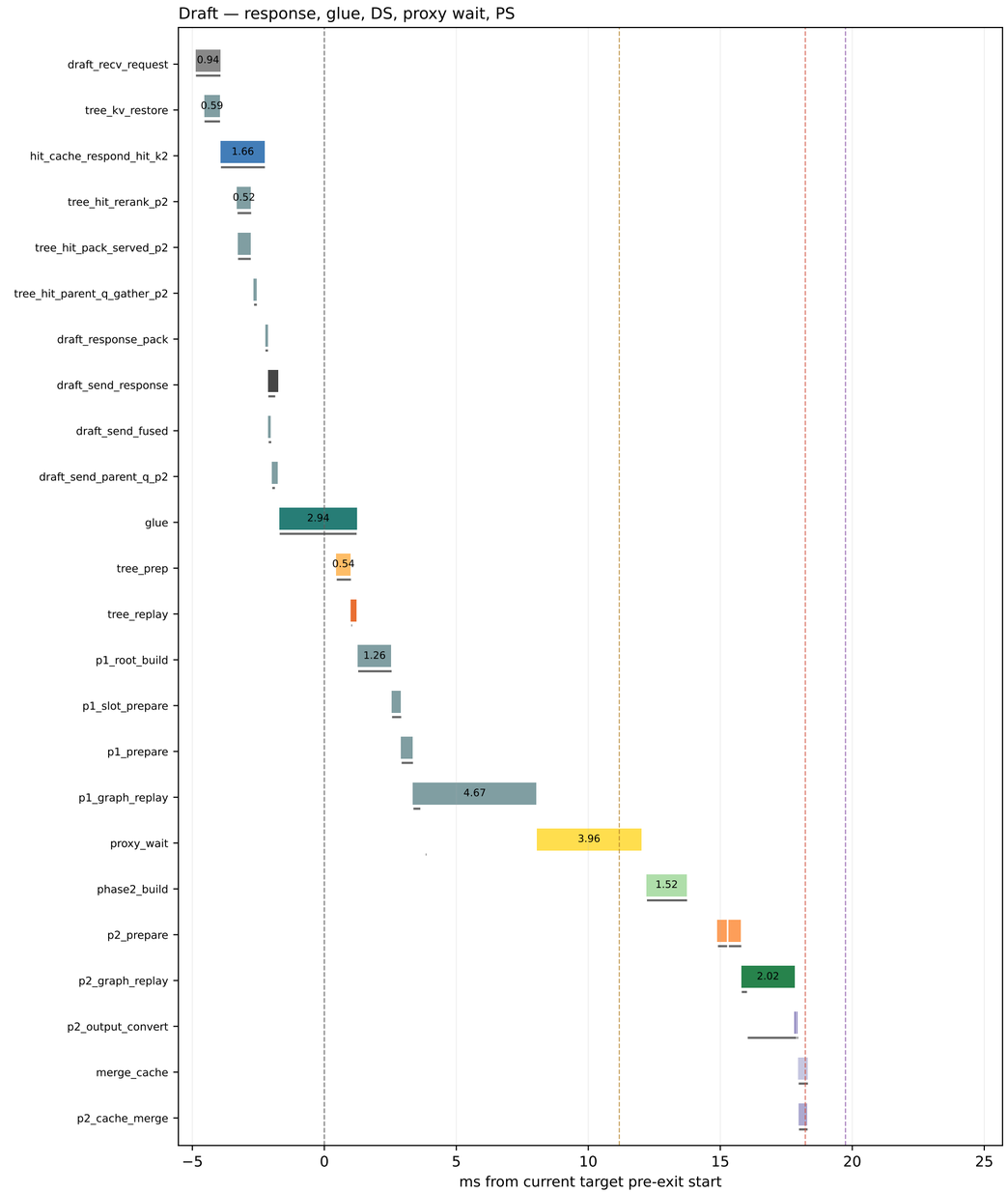
LayerSkip Llama2-7B + llama-68M
P2 hit / p25 target-step duration / request 1, UID 84_t0, step 3

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=4; DS margin +3.959 ms (estimated); PS margin -0.085 ms; next outcome P1 hit





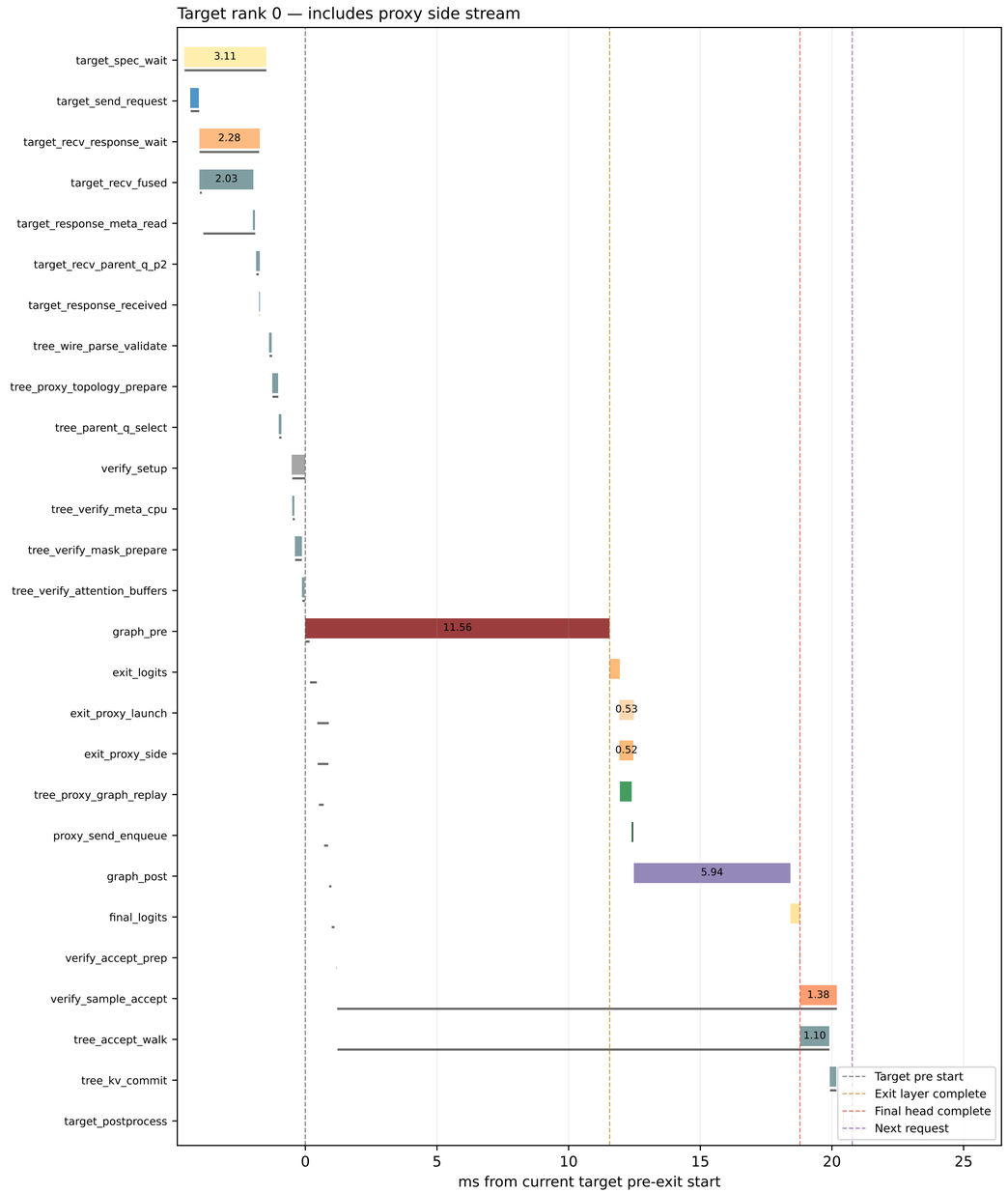
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

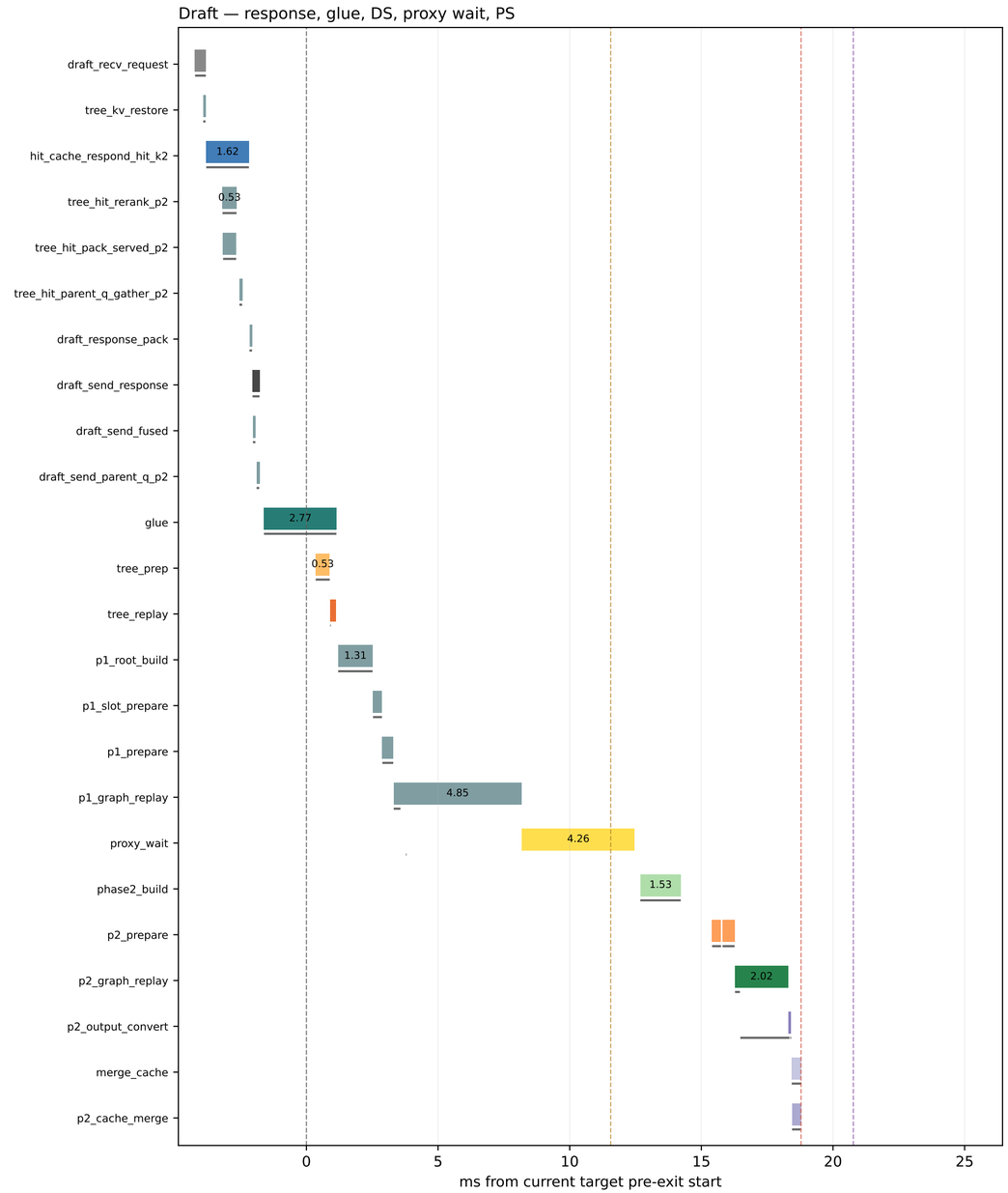
LayerSkip Llama2-7B + llama-68M
P2 hit / p50 target-step duration / request 1, UID 84_t0, step 96

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=3; DS margin +4.265 ms (estimated); PS margin -0.011 ms; next outcome P1 hit





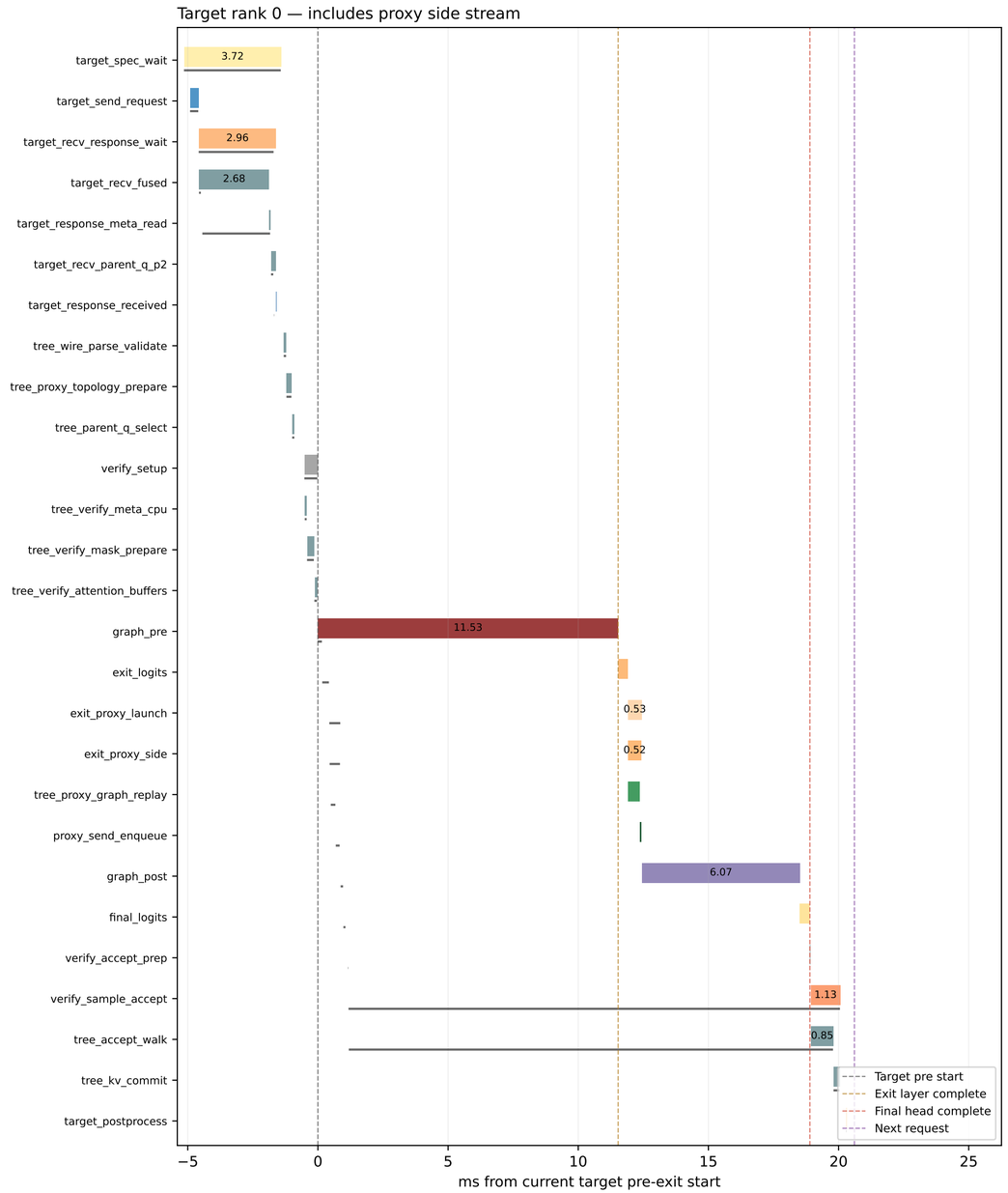
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

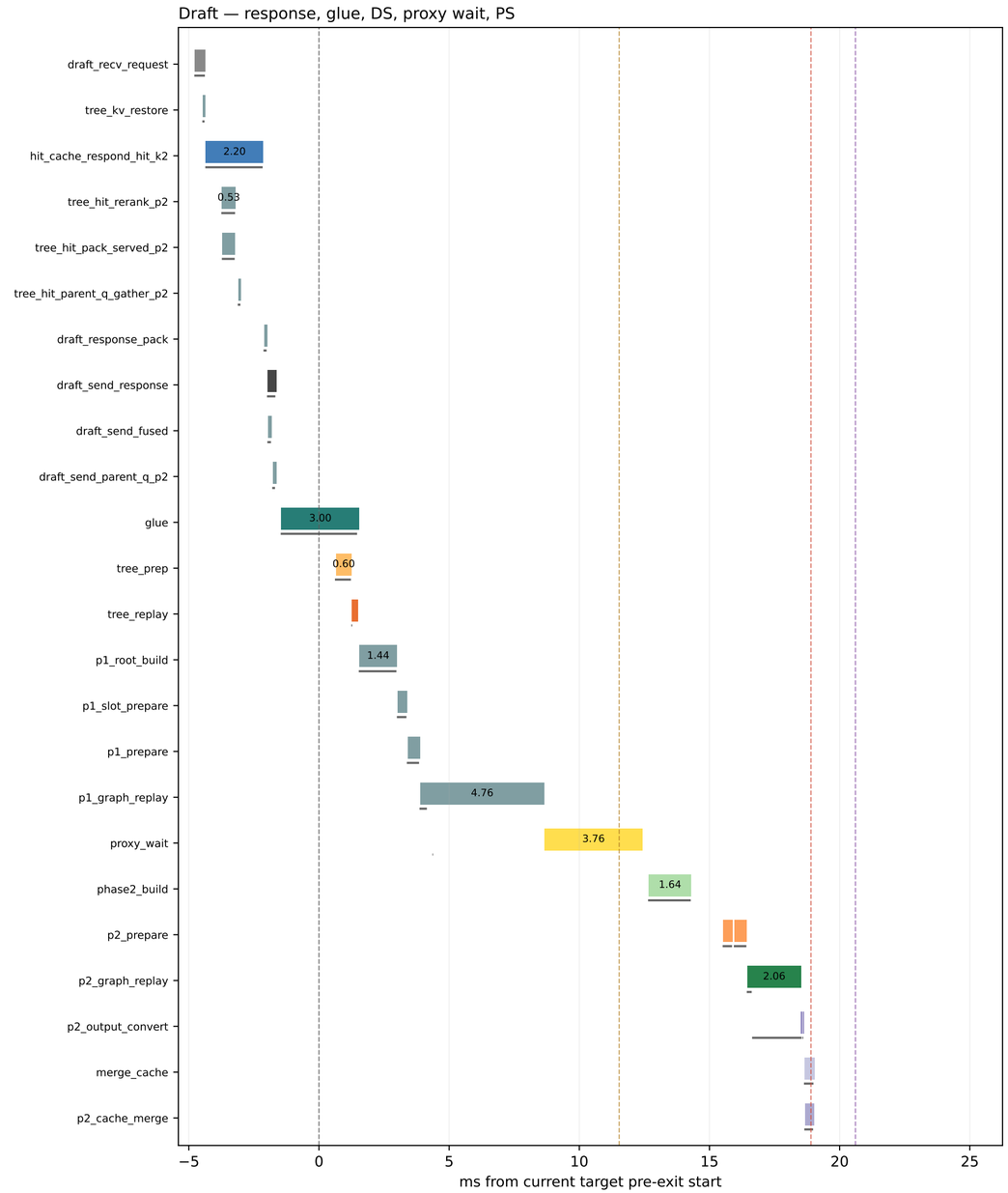
LayerSkip Llama2-7B + llama-68M
P2 hit / p75 target-step duration / request 4, UID 288_t0, step 9

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=3; DS margin +3.763 ms (estimated); PS margin -0.128 ms; next outcome P1 hit





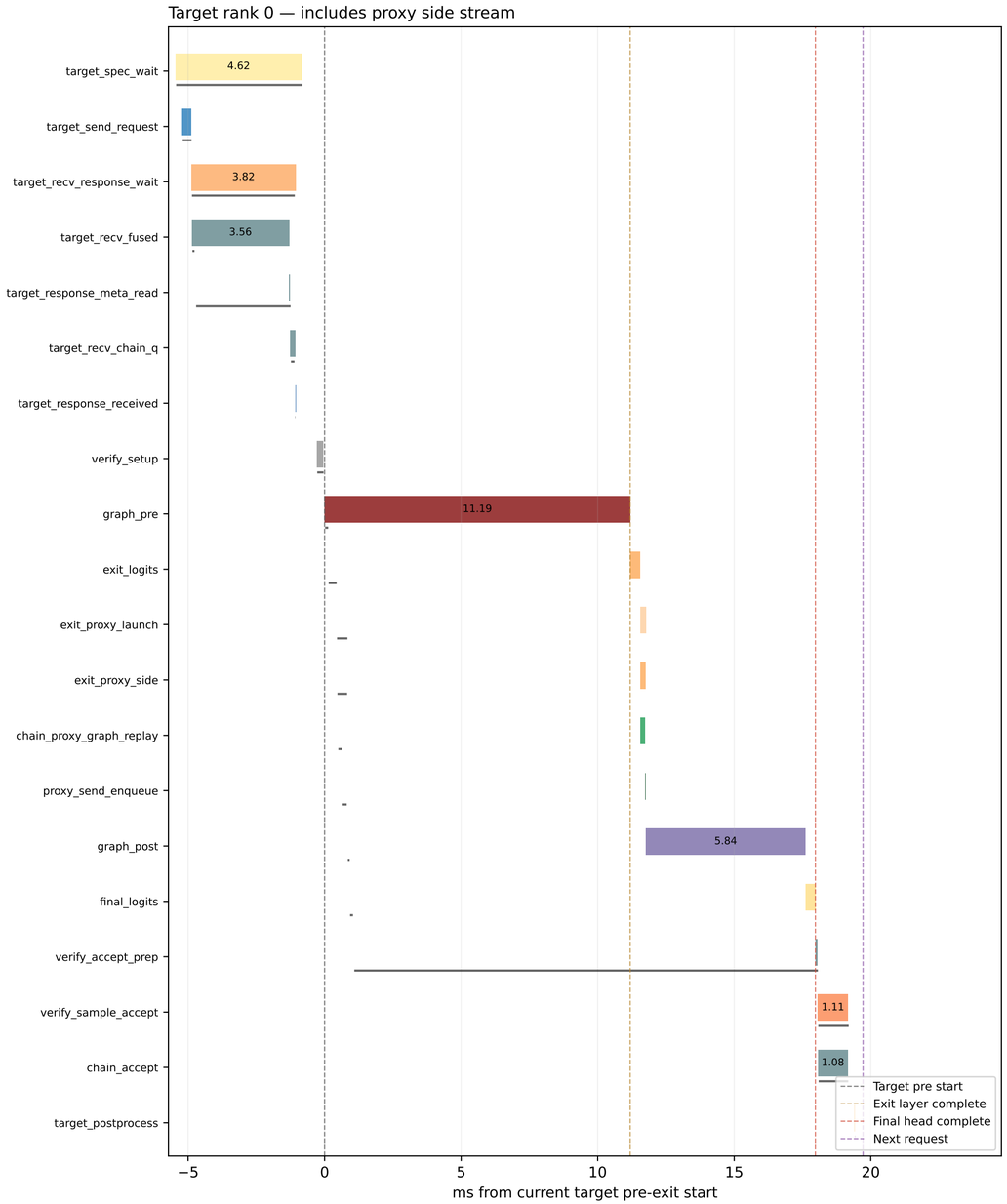
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

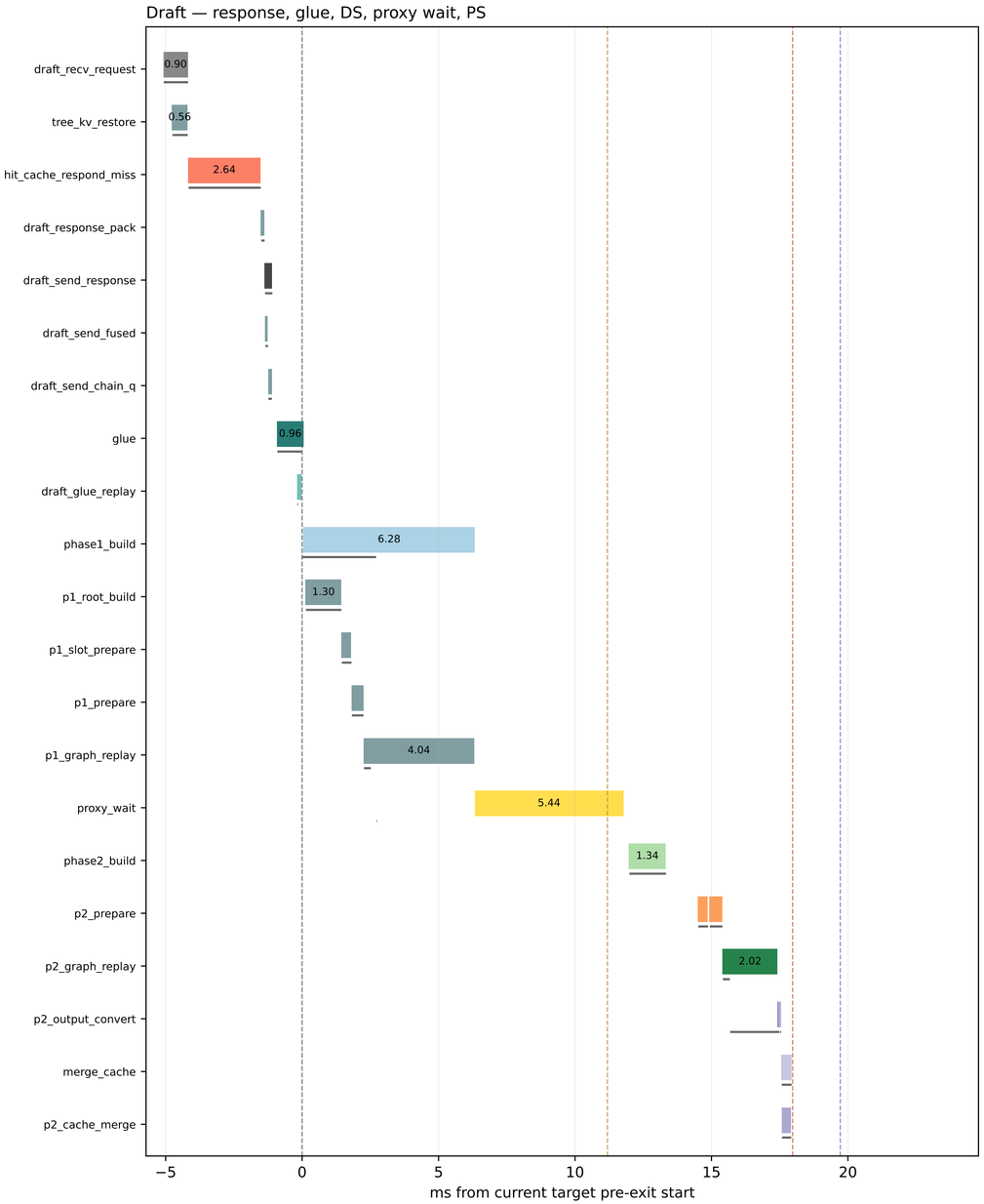
LayerSkip Llama2-7B + llama-68M
Miss / p25 target-step duration / request 1, UID 84_t0, step 16

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=3; DS margin +5.439 ms (estimated); PS margin +0.049 ms; next outcome P2 hit





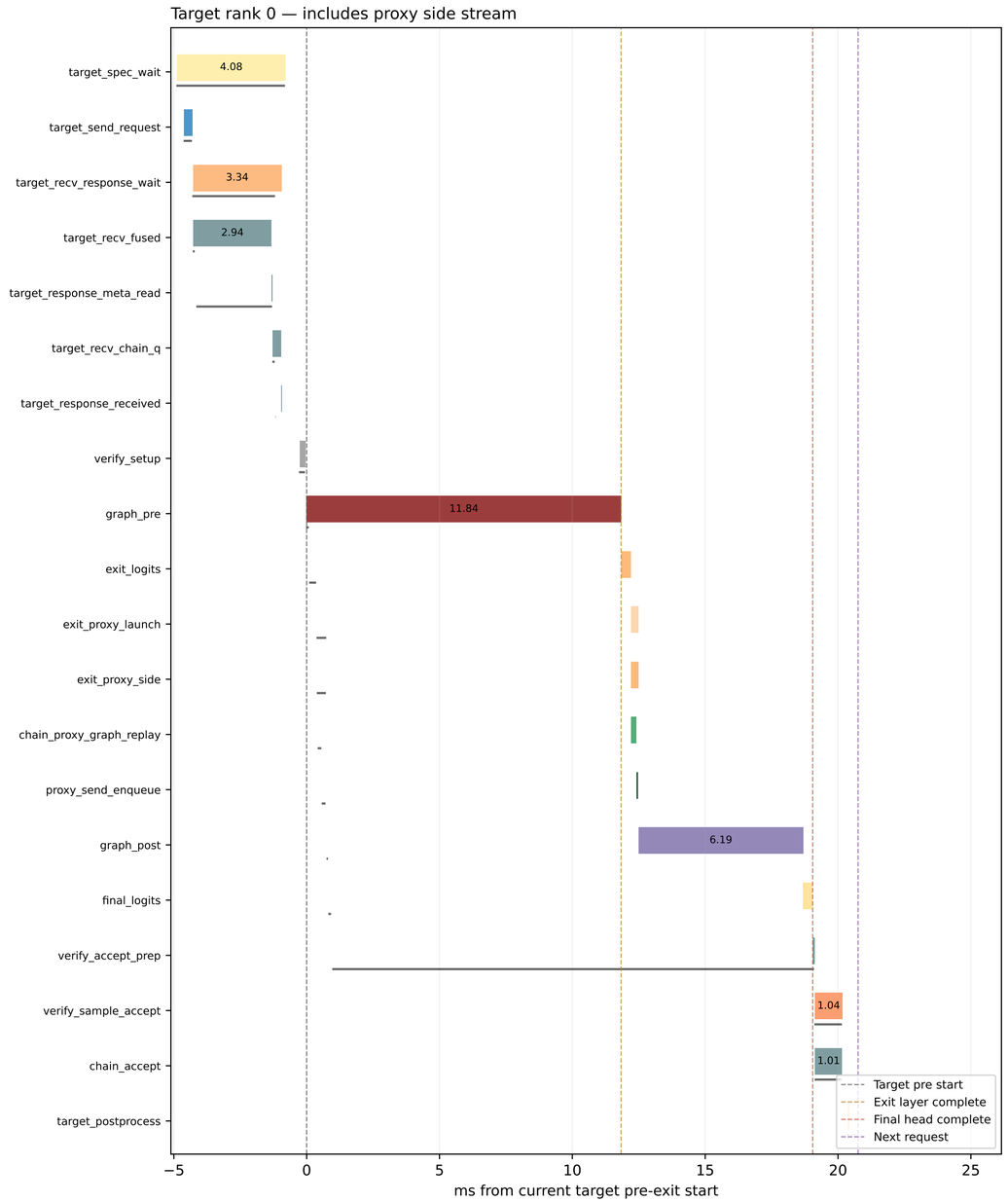
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

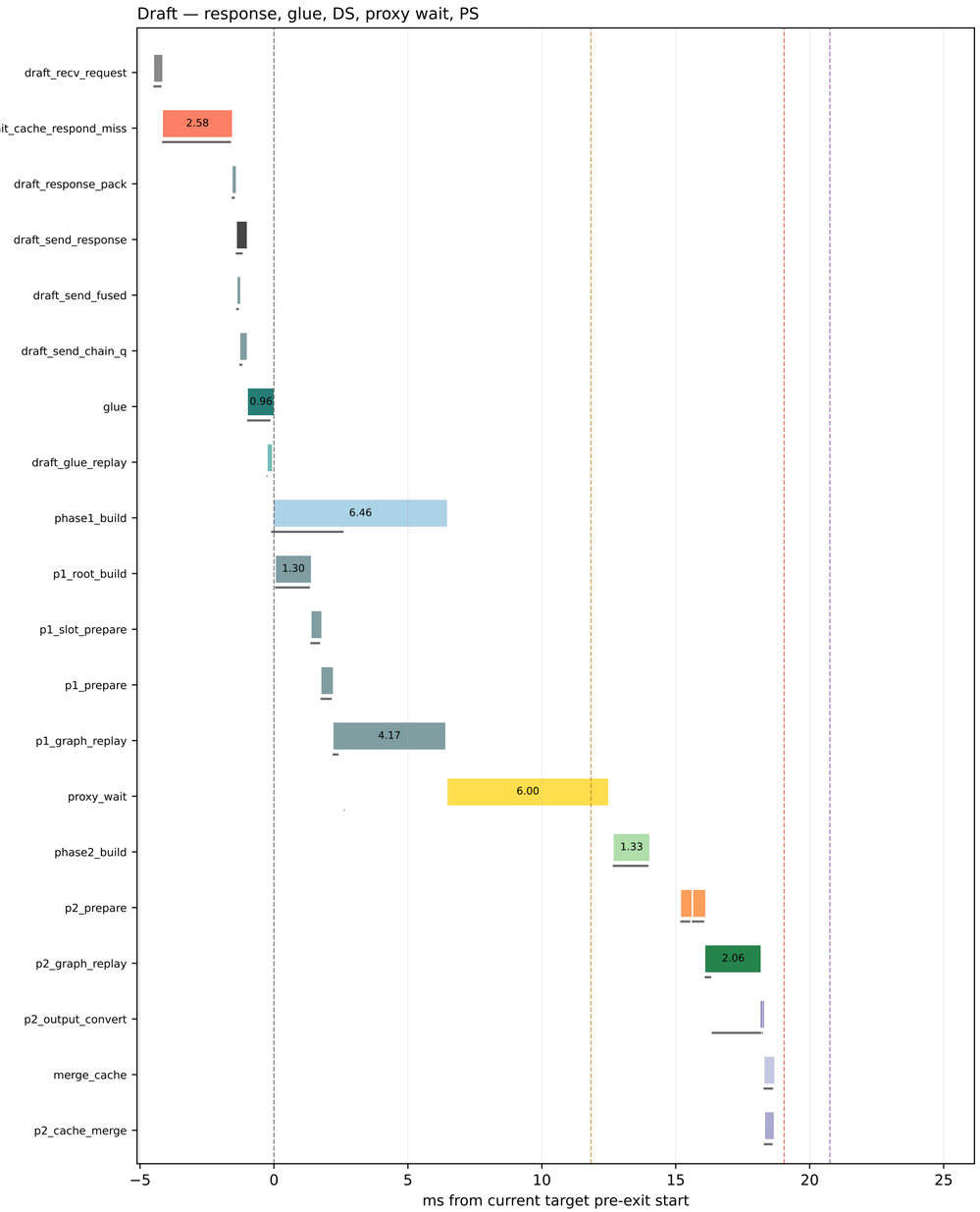
LayerSkip Llama2-7B + llama-68M
Miss / p50 target-step duration / request 4, UID 288_t0, step 84

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +5.994 ms (estimated); PS margin +0.379 ms; next outcome P1 hit





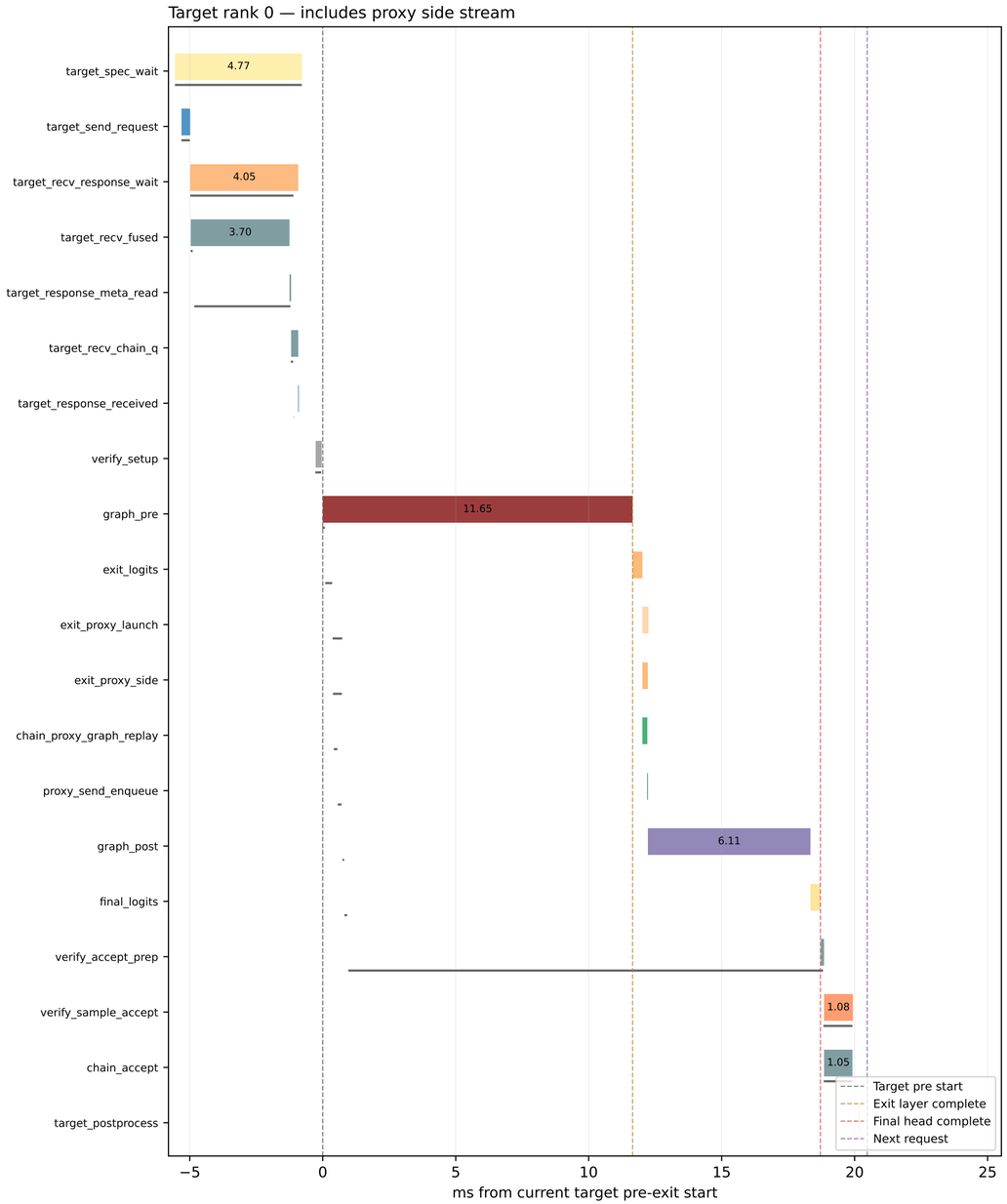
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

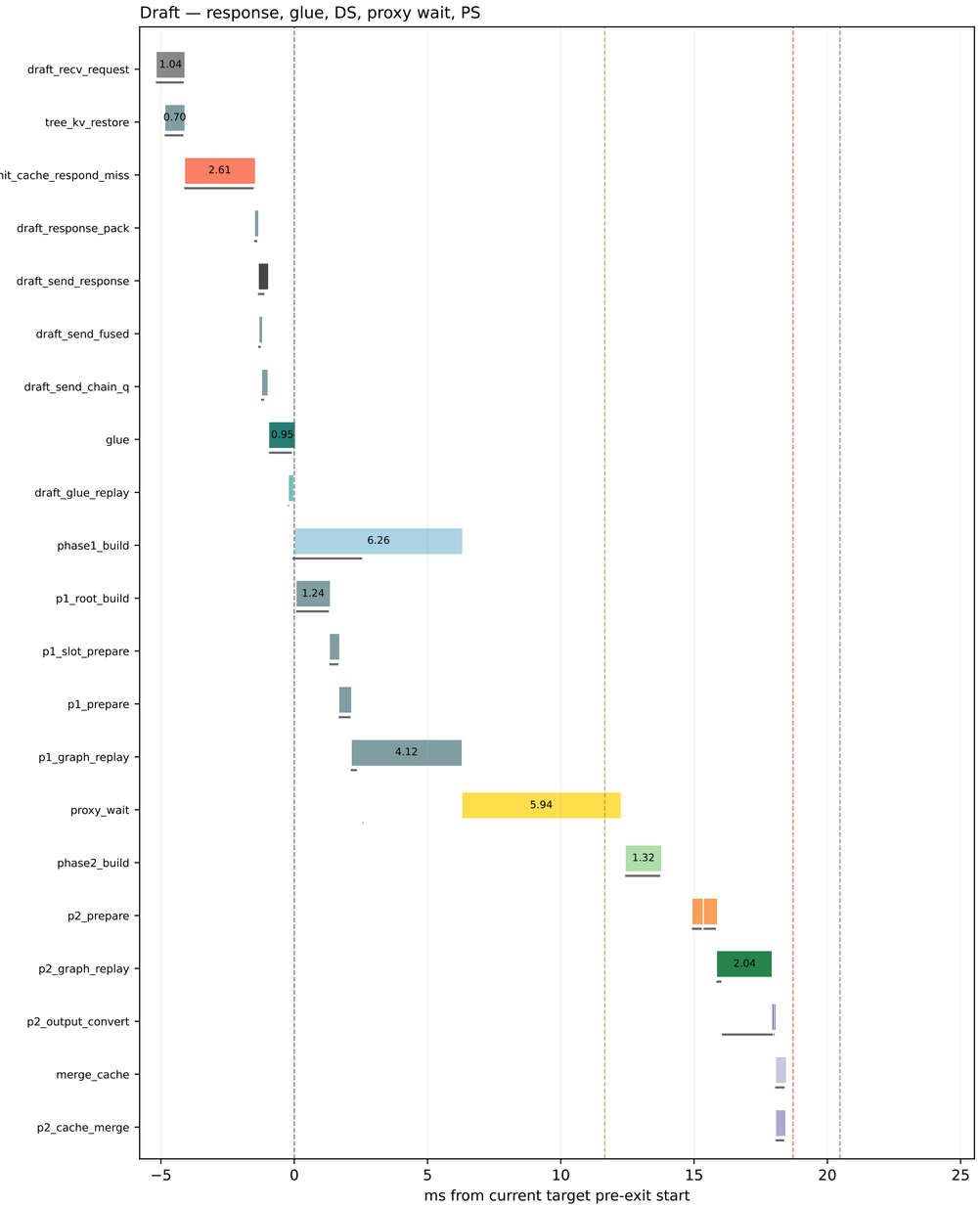
LayerSkip Llama2-7B + llama-68M
Miss / p75 target-step duration / request 4, UID 288_t0, step 38

exit 20 (0-based) | K1/K2 6/3 | G=M DS/PS 12/8 | C=3 | frontier | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 82d964e0 / eslab18 RTX3090 / 4 profiled turns, 321 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +5.941 ms (estimated); PS margin +0.281 ms; next outcome P2 hit





Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.